

DR. CHAO YAO

SENIOR ROBOTICS ENGINEER SPECIALIZING IN AUTONOMOUS NAVIGATION, ROBOTICS INTEGRATION, AND DEVOPS. WITH A STRONG AERIAL ROBOTICS/CONTROL BACKGROUND.

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🎓 Education

Technische Universität Dresden (University of Excellence)

Germany

Dr.-Ing. Robotics

Mar. 2016 – Jul. 2022

- Ph.D thesis:
A Contribution to the Design of Highly Redundant Compliant Aerial Manipulation Systems
- Supervisor: Prof. Dr. techn. Klaus Janschek
- Grade: Magna cum laude

Dipl.-Ing. Electrical Engineering

Oct. 2011 – Jun. 2015

- Thesis: Edge-based Pose Estimation of Known Objects Using Mono-camera (score: **1,0**, poster)

Wuhan University of Science and Technology

China

B.Sc Automation Engineering

Sep. 2006 – Jul. 2010

- Thesis: Autonomous racing car design based on electromagnetic navigator (video)

B.B.A Bussiness Administration

Sep. 2007 – Jul. 2010

- Thesis: Equipment Maintenance Strategy for Wuhan Iron and Steel Corp.

✂ Skills

Robotics & Autonomy: ROS 2; Nav2; Behavior Trees; SLAM; VIO; path planning; exploration; sensor fusion; robot modeling and control

Robotics Software: PX4; Fast-LIO; GBPlanner; Fast-Planner; Bonxai; Basalt; GTSAM; VINS; Ceres Solver; OpenCV

Software Engineering & DevOps: IaC; Docs as Code; Docker; version control; CI/CD; code quality; automated testing; release automation

Robotics Integration: Deployment and adaptation of open-source robotics stacks on x86/ARM; hardware/software bring-up; system integration; performance tuning; troubleshooting

Programming & Scripting: C++; Python; Bash; MATLAB

Simulation: Gazebo; PX4 SITL; Isaac Sim; Flightmare; Simulink

Embedded & Robotics Platforms: NVIDIA Jetson; Intel NUC; Pixracer; Pixhawk; Raspberry Pi; custom MCU boards

R&D: Scientific writing; clean code; agile development; technical leadership

Operating Systems: Linux; Pantavisor; Windows; macOS

Project Management: Scrum; Jira; Confluence; Miro; Polarion

Languages: Chinese (native); German (professional); English (professional)

Driver License: EU; China; UAE

Experience

Senior Robotics SW Engineer

Autonomous Robotics Research Center, Technology Innovation Institute

- Founder of team DevOps, Docker orchestration and QA framework
- Architect of the indoor navigation software stack
- Lead of BehaviorTree, lifecycle management
- Expert of robotic software integration

Jun. 2024 – present

Abu Dhabi, UAE

Robotics Software Developer

Home Care Robotics, BSH Home Appliance GmbH (BOSCH subsidiary)

- Expert of robotic software for navigation & behavior
- Specialist of connectivity integration for Home Connect

Jan. 2022 – May. 2024

Bad Neustadt, Germany

Teaching & research assistant

Institute of Automation, TU Dresden

- Associate Lecturer of "Serial Manipulator Control" and "Mobile Robot Control"
- Teaching assistant of "Automation Technology"
- Supervisor of student projects (15+ master/diploma theses, 20+ bachelor/student theses)
- Lead of robotic laboratory
- Associate Lecturer of annual summer school for School of Mechanical Engineering, Zhejiang University
- Researcher of teaching and research projects

Aug. 2015 – Oct. 2021

Dresden, Germany

Session chair

WROCO 2019 - The 1st IFAC Workshop on Robot Control

Sep. 2019

Daejeon, Korea

Research visit

Department of Automation, East China University of Science and Technology

- Visiting scholar in the robotics laboratory led by Prof. Yang Tang

Sep. 2017

Shanghai, China

Internship

ADC Automotive Distance Control Systems GmbH (Continental AG)

- Programmed MS Word and Access for automatic import of SAP data in local database
- Developed programs to run endurance HIL test for Radar sensors with CANoe and Matlab

Feb. 2014 – Aug. 2014

Lindau am Bodensee, Germany

National university students competition

Smart Car Laboratory, Wuhan University of Science and Technology

- Developed a camera-based autonomous model car in a team of 3
- Lead of navigation, control and electronics

Mar. 2009 – Aug. 2009

Wuhan, China

Projects

Crane Interior Inspection Drone | *ROS2, Fast-LIO, PX4, Jetson*

- Own DevOps and software architecture
- Lead ROS2 Software development
- Safety pilot

Oct. 2025 – present

Aerial RFID Scanning for Warehouses | *Fast-LIO, Fast-Planner, PX4, Jetson*

- Led software integration
- Led DevOps
- executed field tests

Jun. 2024 – Oct. 2025

BOSCH Cleaning Robot 2 | *Nav2, Grid Map, Gazebo, D-Bus, Jetson*

- Led robot integration into a smart home devices ecosystem
- Developed robotic navigation & behavior software

Jan. 2022 – Jul. 2023

Flying Manipulator | *ROS, Gazebo, Matlab/Simulink, CAD, NUC* Jun. 2018 – Sep. 2021

- Built a fully actuated UAV prototype
- Derived mathematical models for a high-dimensional aerial manipulation system
- Developed Matlab & Gazebo simulators
- Nonlinear controllability and control authority analysis
- Designed & evaluated various controllers and observers
- Designed an active fault-tolerant control scheme for rotor faults

Teaching Robots in Saxony | *ROS, Gazebo, Matlab/Simulink* Aug. 2015 – Mar. 2017

- Integrated advanced hardware for complex tasks
- Developed tools for joint controller tuning
- Developed cartesian space control with a demonstration of drawing the university logo
- Developed SLAM and Guidance-Navigation-Control stack for a mobile robot

Image Recognition and Processing for Navigation | *Opencv, Matlab* Dec. 2014 – May 2015

- contributed to the CV algorithms for visible spectrum camera in the navigation function for European Space Agency
- Developed algorithms for relative pose estimation on a scenario with uncooperative targets in an Active Debris Removal configuration

Kleine Linear- und Planardirektantriebe | *Altium Designer, Matlab* Jun. 2013 – Sep. 2013

- configured and verified the position measuring system based on optical encoder and interpolator for a precision linear drive
- designed a position controller based on state feedback and Luenberger observer for a planar drive

Research

Research interests

Aerial Robotics Over-actuated Systems Fault-tolerant Control Observer Design
Nonlinear Controllability Visual Inertial Odomtry SLAM 3D Path Planning

Publications (✉ see  ORCID for a list)

Awards

Best Poster Award Mar. 2019
VDI Mechatronik 2019 Paderborn, Germany

- Wrench space based design of fully actuated aerial robots

Best Bachelor Thesis Jul. 2010
of Hubei Province (scope: 130 colleges and universities) Wuhan, China

- Autonomous racing car design based on electromagnetic navigator

Second prize of the National Final Aug. 2009
National University Students Intelligent Car Race Beijing, China

First prize of the Qualifier South China Jul. 2009
National University Students Intelligent Car Race Guangzhou, China